

Contents

BioDYM — Getting Started From Zero	1
What you will install	1
Step 1 — Install VS Code	1
Add the Python & Jupyter extensions (recommended)	2
Step 2 — Install uv (this also gives you Python)	2
Step 3 — Get the BioDYM project onto your computer	2
Option A — Use Git (recommended — makes updates painless)	2
Option B — Download as a ZIP (quick one-time look, no updates)	3
Step 4 — Install BioDYM’s environment	3
Step 5 — Run BioDYM	3
A) The dashboard — point-and-click, zero coding (best first experience)	3
B) The analysis notebook — the full, guided workflow	4
C) The SystemDefiner — build your own system visually (optional)	4
Updating BioDYM later	4
Everyday cheat sheet	4
Troubleshooting	5
Quick recap	5

BioDYM — Getting Started From Zero

A complete, no-prior-knowledge guide to installing everything you need and running BioDYM for the first time. **You do not need to know how to program.** Just follow the steps in order.

What is BioDYM? A tool that analyses how biological materials (straw, compost, organic waste, carbon...) flow and accumulate through a system over time. You give it data (an Excel file or a config you build in a visual editor), and it produces charts, mass balances, and tables.

Estimated time: 30–45 minutes, most of it waiting for downloads.

What you will install

You only need **two** things. One tool (**uv**) automatically handles the rest (Python, Jupyter, and ~30 scientific libraries) for you — so don’t install Python separately.

Tool	What it is	Why you need it
VS Code	A free code/text editor from Microsoft	To open the project and run a terminal
uv	A fast Python project manager	It installs the correct Python, Jupyter, and all libraries automatically

You may have heard you need to “install Python” and “install Jupyter” yourself. With **uv** you **don’t** — it downloads the exact Python version BioDYM needs and every library, all into a self-contained folder inside the project. Nothing is installed system-wide, so it can’t break other software on your computer.

Step 1 — Install VS Code

1. Go to <https://code.visualstudio.com>
2. Click the big download button (it auto-detects Windows/Mac).

3. Run the installer. On Windows, **tick “Add to PATH”** if asked (it usually is by default). Accept the defaults for everything else.
4. Launch VS Code once to confirm it opens.

Add the Python & Jupyter extensions (recommended)

Inside VS Code:

1. Click the **Extensions** icon in the left bar (four squares) — or press **Ctrl+Shift+X**.
2. Search for **“Python”** (by Microsoft) → click **Install**.
3. Search for **“Jupyter”** (by Microsoft) → click **Install**.

These let you open notebooks directly inside VS Code.

Step 2 — Install uv (this also gives you Python)

uv is installed with a single command. You’ll use VS Code’s built-in terminal.

1. In VS Code, open the terminal: menu **Terminal** → **New Terminal** (or press **Ctrl+`** — the backtick key, top-left of most keyboards).
2. Copy the command for your system, paste it into the terminal, and press **Enter**.

On Windows (PowerShell — this is what VS Code opens by default on Windows):

```
powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/install.ps1 | iex"
```

On macOS / Linux:

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

3. **Close the terminal and open a new one** (Terminal → New Terminal). This is important — it lets your computer “find” the newly installed uv.
4. Verify it worked by typing:

```
uv --version
```

You should see something like `uv 0.5.x`. If you get “command not found”, restart VS Code completely and try again.

You never have to install Python yourself. When you run `uv sync` in Step 4, uv reads BioDYM’s requirements, downloads the right Python version automatically, and sets everything up.

Step 3 — Get the BioDYM project onto your computer

Pick **one** of the two options below.

Which should I choose? If you expect to receive updates to BioDYM (almost everyone does), use **Option A (Git)**. Updating then takes two commands instead of re-downloading and re-unzipping the whole project — and you won’t lose any data or config you added to the folder. Only use the ZIP (Option B) for a quick one-time look.

Option A — Use Git (recommended — makes updates painless)

First install Git (a small tool that downloads the project and later fetches updates):

1. Go to <https://git-scm.com/downloads> and download Git for your system.
2. Run the installer. **Accept every default** — there are several screens, just keep clicking *Next*. (The defaults are correct for BioDYM; you don’t need to understand the options.)

3. **Fully close and reopen VS Code** so it can find the newly installed Git.

Then download the project. In a VS Code terminal (**Terminal** → **New Terminal**), run:

```
git clone https://github.com/JScholz-tech/Biodym_JS.git
```

This creates a `Biodym_JS` folder in whatever location the terminal is pointing at (by default your user folder). Then in VS Code: **File** → **Open Folder...** → select that new `Biodym_JS` folder.

Verify Git works: run `git --version` in the terminal. If you see a version number, you're good. "command not found" usually means you need to fully restart VS Code.

Option B — Download as a ZIP (quick one-time look, no updates)

1. Go to the project page: https://github.com/JScholz-tech/Biodym_JS
2. Click the green < > **Code** button → **Download ZIP**.
3. Unzip it somewhere easy to find, e.g. your **Documents** folder.
4. In VS Code: **File** → **Open Folder...** and select the unzipped `Biodym_JS` folder.

A ZIP is a frozen snapshot. To get a newer version later you'd download and unzip again from scratch — which is exactly why Option A is recommended.

You're "in" the project when the VS Code title bar / Explorer panel shows the project files (`00_BioDYM_Workflow.ipynb`, `pyproject.toml`, the `02_src` folder, etc.) and the terminal prompt is inside that folder.

Step 4 — Install BioDYM's environment

With the project folder open in VS Code, open a terminal (**Terminal** → **New Terminal**) and run:

```
uv sync
```

What this does: downloads the correct Python, creates a private `.venv` folder inside the project, and installs Jupyter plus every scientific library BioDYM needs.

This is the longest step — it downloads a few hundred MB the first time. Wait until you get your terminal prompt back. (Future runs are nearly instant because everything is cached.)

Golden rule: From now on, always start your commands with `uv run`. That prefix tells your computer to use the project's private environment instead of hunting for a system Python that may not exist. `uv run jupyter ...`, `uv run python ...`, and so on.

Step 5 — Run BioDYM

There are three ways to use BioDYM. If you're brand new, start with **A**.

A) The dashboard — point-and-click, zero coding (best first experience)

```
uv run voila 01_BioDYM_Dashboard.ipynb
```

A tab opens in your web browser with interactive charts and controls. When you're done, return to the terminal and press `Ctrl+C` to stop it.

B) The analysis notebook — the full, guided workflow

```
uv run jupyter lab
```

Your browser opens **Jupyter Lab**. In the file list on the left, double-click `00_BioDYM_Workflow.ipynb`. Then in the top menu choose **Kernel** → **Restart Kernel and Run All Cells**. The notebook runs top to bottom and produces all the charts and tables.

- It already points at a working example file, so it runs out of the box.
- To use your own data, find the cell near the top that sets `input_file = ...` and change it to your own Excel file or `config.yaml`, then run all cells again.

When finished, go back to the terminal and press **Ctrl+C** to shut Jupyter down.

C) The SystemDefiner — build your own system visually (optional)

If you want to design your own system without editing Excel by hand:

```
uv run python -m systemdefiner
```

Then open <http://localhost:8001> in your browser. Lay out your processes and flows, and export a `config.yaml` that you can feed into the notebook (option B). Press **Ctrl+C** in the terminal to stop it.

Updating BioDYM later

This works only if you got the project with **Git** (Step 3, Option A). ZIP users have to download a fresh ZIP instead.

When a newer version of BioDYM is released, open the project folder in VS Code, open a terminal, and run these **two** commands:

```
git pull
uv sync
```

- `git pull` downloads the latest code changes.
- `uv sync` installs any new or updated libraries the new version needs.

That's the whole update. Your own files in `01_data/` (Excel inputs, outputs, your `config.yaml`) are left untouched.

If `git pull` reports a conflict (because you edited a tracked file), the simplest fix is to copy your changed file somewhere safe, run `git checkout -- <that file>` to discard the local change, then `git pull` again. If unsure, ask whoever maintains the project rather than guessing.

Everyday cheat sheet

Once installed, this is all you need day to day. Open the project folder in VS Code, open a terminal, and run:

I want to...	Command
Open the click-through dashboard	<code>uv run voila</code> <code>01_BioDYM_Dashboard.ipynb</code>
Open the full analysis notebook	<code>uv run jupyter lab</code>
Build a system visually	<code>uv run python -m systemdefiner</code> → <code>http://localhost:8001</code>
Update BioDYM to the latest version (Git only)	<code>git pull</code> then <code>uv sync</code>
Stop any running tool	Press Ctrl+C in the terminal

I want to...	Command
--------------	---------

Troubleshooting

uv is not recognized / “command not found” You opened the terminal before installing `uv`, or didn’t restart it. Fully close VS Code, reopen it, open a new terminal, and try `uv --version` again.

uv sync fails or seems stuck Check your internet connection (it downloads a lot the first time). If it errored, just run `uv sync` again — it safely resumes.

A command “can’t find” Python or a library You probably forgot the `uv run` prefix. `python xyz` won’t work; `uv run python xyz` will.

The browser tab didn’t open by itself Look in the terminal output for a line with a web address (e.g. `http://localhost:8888/...` or `http://localhost:8001`) and paste it into your browser manually.

“Port already in use” A previous run is still going. Find that terminal and press `Ctrl+C`, or just close all terminals and start fresh.

I edited a .py file in 02_src/ but the notebook didn’t change Restart the notebook’s kernel: **Kernel → Restart Kernel and Run All Cells**. BioDYM does not auto-reload changed code.

git is not recognized You ran a `git` command before installing Git, or didn’t restart VS Code afterward. Install Git (Step 3, Option A), fully close and reopen VS Code, then try `git --version`.

Still stuck? See the detailed [README](#), or open an issue at https://github.com/JScholz-tech/Biodym_JS/issues.

Quick recap

1. Install **VS Code** → add **Python + Jupyter** extensions.
2. Install **uv** (one command) — it brings Python with it.
3. Install **Git**, then `git clone` the project and open the folder in VS Code. (*Or download a ZIP for a one-time look.*)
4. Run `uv sync` once.
5. Run `uv run voila 01_BioDYM_Dashboard.ipynb` (or `uv run jupyter lab`).
6. Later, update anytime with `git pull` then `uv sync`.

That’s it — you’re running BioDYM.